



Lista-5



A partir das equações governantes do estado média da atmosfera.

$$\frac{\partial U}{\partial t} + \frac{1}{a \cos^2 \theta} \left(U \frac{\partial U}{\partial \lambda} + V \cos \theta \frac{\partial U}{\partial \theta} \right) + \dot{\eta} \frac{\partial U}{\partial \eta} - fV + \frac{1}{a} \left(\frac{\partial \Phi}{\partial \lambda} + R_d T_v \frac{\partial \ln p}{\partial \lambda} \right) = F_u \quad (1)$$

$$\frac{\partial V}{\partial t} + \frac{1}{a \cos^2 \theta} \left(U \frac{\partial V}{\partial \lambda} + V \cos \theta \frac{\partial V}{\partial \theta} \right) + \dot{\eta} \frac{\partial V}{\partial \eta} + fU + \frac{\cos \theta}{a} \left(\frac{\partial \Phi}{\partial \theta} + R_d T_v \frac{\partial \ln p}{\partial \theta} \right) + \frac{\sin \theta}{a \cos^2 \theta} (U^2 + V^2) = F_v \quad (2)$$

$$\frac{\partial T}{\partial t} + \frac{1}{a \cos^2 \theta} \left(U \frac{\partial T}{\partial \lambda} + V \cos \theta \frac{\partial T}{\partial \theta} \right) + \dot{\eta} \frac{\partial T}{\partial \eta} - \frac{\kappa T_v \omega}{(1 + (\delta - 1)q)p} = F_T \quad (3)$$

$$\frac{\partial q}{\partial t} + \frac{1}{a \cos^2 \theta} \left(U \frac{\partial q}{\partial \lambda} + V \cos \theta \frac{\partial q}{\partial \theta} \right) + \dot{\eta} \frac{\partial q}{\partial \eta} = F_q \quad (4)$$

$\theta = \text{latitude}$
 $\lambda = \text{Longitude}$
 $U = \text{Velocidade Zonal}$

$$\frac{\partial \ln p_s}{\partial t} + \frac{1}{p_s} \int_0^1 \nabla \cdot \left(\mathbf{v}_H \frac{\partial p}{\partial \eta} \right) d\eta = 0 \quad (5)$$

$$\frac{\partial \phi}{\partial \eta} + \frac{R_d T_v}{p} \frac{\partial p}{\partial \eta} = 0 \quad (6)$$



F_u = *tendencia de momemtum zonal das parametrizações Fisica*

F_v = *tendencia de momemtum zonal das parametrizações Fisica*

F_T = *tendencia de temperatura das parametrizações Fisica*

F_q = *tendencia de temperatura das parametrizações Fisica*

Neste modelo representada pelo temo de relaxamento Newtoniano (Amortecimento)

$$F_x = Tend_x = \frac{\Psi_{model}^n - \Psi_{referency}^n}{\tau_{scala\ temporal\ de\ relaxamento}}$$

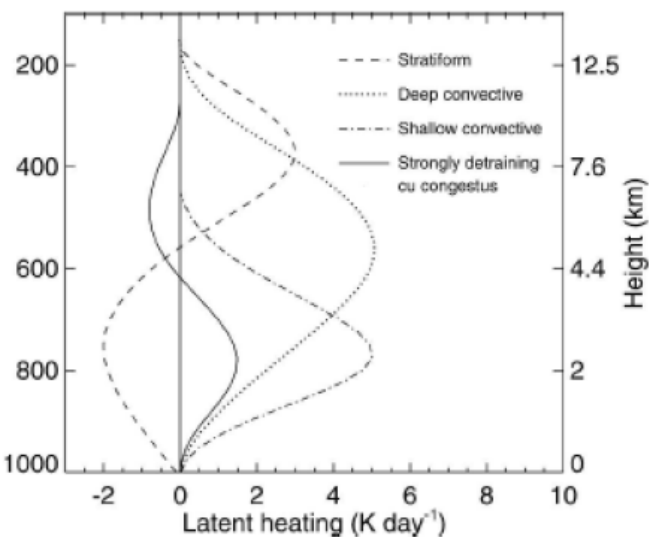


FIG. 1. Idealized latent heating profiles for different precipitating cloud types.

IR and solar radiative heating/cooling rates of clouds with different vertical distribution:

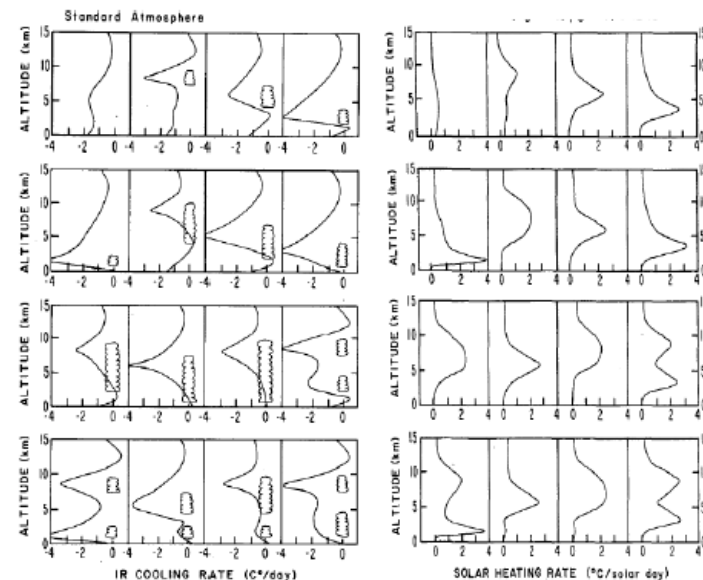
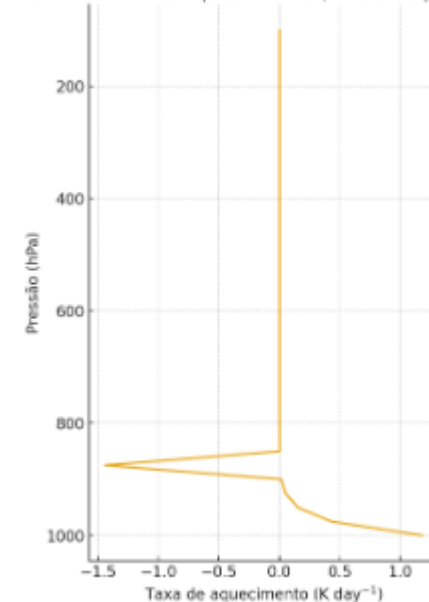


Figure 21.2 IR and solar heating/cooling rate profiles for clear and various cloudy atmospheric conditions. 100% cloud cover is assumed. Solar heating rates were calculated for a solar constant of 1360 W/m^2 , surface albedo of 0.15, cosine of solar zenith angle 0.5 and solar time duration of 12 hours (modified from Liou and Zheng, 1984).

Perfil teórico de aquecimento (níveis de pressão)



$$F_T = \bar{Q}_1^{total} = \left(\frac{\partial T}{\partial t} \right)_{rad} + \left(\frac{\partial T}{\partial t} \right)_{conv} + \left(\frac{\partial T}{\partial t} \right)_{micr} + \left(\frac{\partial T}{\partial t} \right)_{pbl} + \left(\frac{\partial T}{\partial t} \right)_{super}$$

$$F_q = \bar{Q}_2^{total} = \left(\frac{\partial q}{\partial t} \right)_{rad} + \left(\frac{\partial q}{\partial t} \right)_{conv} + \left(\frac{\partial q}{\partial t} \right)_{micr} + \left(\frac{\partial q}{\partial t} \right)_{pbl} + \left(\frac{\partial q}{\partial t} \right)_{super}$$



1) Implemente este modelo baseado nas **Equações do Estado Média Da atmosfera**

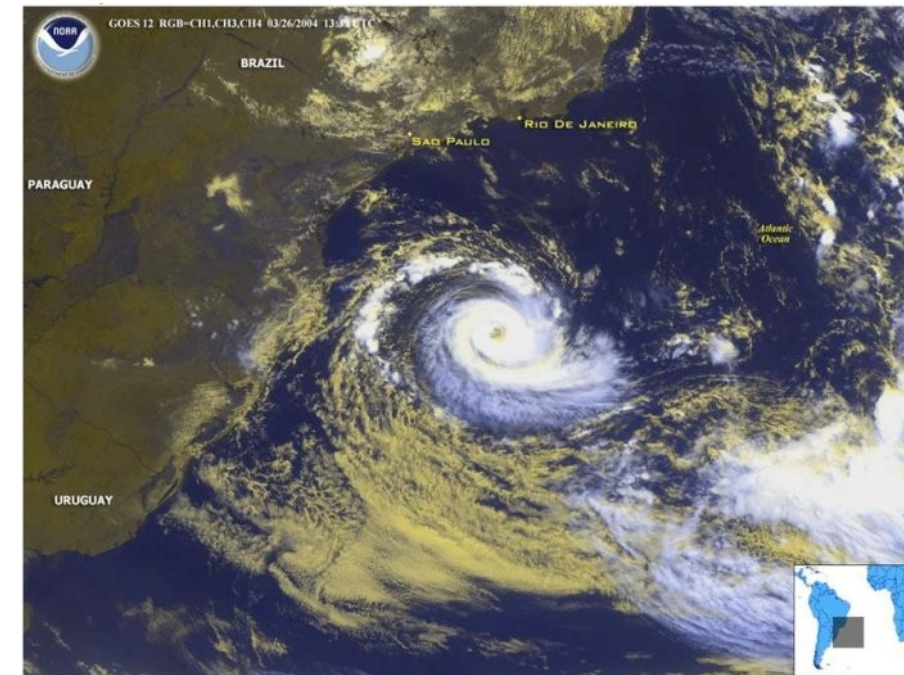
https://github.com/pkubota/pgmet_2025/tree/main/MET-348-3/scm_dynamics_2025

2) Integre o modelo para o caso do furacão Catarina e compare os campos de **velocidade do vento**, **Temperatura** e **Umidade Especifica** com os dados do Era5?

Link dos dados do furacão Catarina

https://ftp.cptec.inpe.br/pesquisa/bam/paulo.kubota/externo/MET-348-3/scm_pgmet_2025.tar.gz

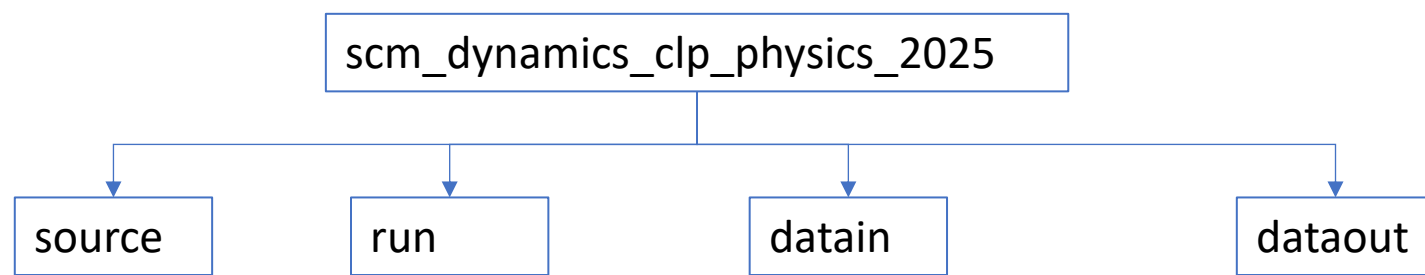
https://ftp.cptec.inpe.br/pesquisa/bam/paulo.kubota/externo/MET-348-3/scm_dynamics_2025/dataout/





No modelo proposto no link

https://github.com/pkubota/pgmet_2025/tree/main/MET-348-3/scm_dynamics_clp_physics_2025



Class_Module_Constants.f90
Class_Module_TimeManager.f90
Class_Module_Dynamics.f90
Class_Module_Fields.f90
Main.f90
namelist.input

https://ftp.cptec.inpe.br/pesquisa/bam/paulo.kubota/externo/MET-348-3/scm_pgmet_2025.tar.gz



Use o namelist e veja a contribuição de cada parametrização modificando as forçantes (ForCLW, ForcSW, ForcDC, ForcPB)

```
&TimeControl
  idatei=00,24,03,2004      ,!
  idatec=00,24,03,2004      ,!
  idatef=23,28,03,2004      ,!
  DHFCT= 24                 ,!
  dt_step= 180.0             ,!
  ICtrDay= 0                 ,!
/
&MeshesControl
  nLon    = 93    !number of longitude !161
  nLat    = 61    !number of latitude  161
  InitLon  = 302.0 !initial longitude 0 - 360
  InitLat  = -35.0 !initial latitude -90  90
  DeltaLon = 0.25 !resolution x-axis
  DeltaLat = 0.25 !resolution y-axis
  nLev     = 37    ! number of vertical layers
  vis      = 2.5e3  ! Coeficiente de difusão para  momentum e temperatura
  vis_q    = 3.0e4  ! Coeficiente de difusão para  umidade
  tau1     = 1800   ! Amortecimento para as forçantes físicas
/
&ForcingControl
  ForCLW = 1  ,!ForCLW = 1 => on or ForCLW = 0 => off  LONG WAVE RADIATIVE HEATING
  ForcSW = 1  ,!ForcSW = 1 => on or ForcSW = 0 => off  SHORT WAVE RADIATIVE HEATING
  ForcDC = 1  ,!ForcDC = 1 => on or ForcDC = 0 => off  CONVECTIVE LATENT HEATING
  ForCLG = 1  ,!ForCLG = 1 => on or ForCLG = 0 => off  LARGE SCALE LATENT HEATING
  ForcSC = 1  ,!ForcSC = 1 => on or ForcSC = 0 => off  SHALLOW CONVECTIVE HEATING
  ForcPB = 1  ,!ForcPB = 1 => on or ForcPB = 0 => off  VERTICAL DIFFUSION HEATING
/
```



Use o namelist e veja a contribuição de cada parametrização modificando as forçantes (ForCLW, ForCSW, ForCDC, ForCPB)

OBS: Lembre-se que o namelist “**namelist.input**” deve estar no diretório que você irá executar o modelo

Objetivo técnico: Compara cada caso (experimentos) com o experimento com todas as forçantes ligadas

```
&ForcingControl  
  ForCLW = 1 , !ForCLW = 1 => on or ForCLW = 0 => off LONG WAVE RADIATIVE HEATING  
  ForCSW = 1 , !ForCSW = 1 => on or ForCSW = 0 => off SHORT WAVE RADIATIVE HEATING  
  ForCDC = 1 , !ForCDC = 1 => on or ForCDC = 0 => off CONVECTIVE LATENT HEATING  
  ForCLG = 1 , !ForCLG = 1 => on or ForCLG = 0 => off LARGE SCALE LATENT HEATING  
  ForCSC = 1 , !ForCSC = 1 => on or ForCSC = 0 => off SHALLOW CONVECTIVE HEATING  
  ForCPB = 1 , !ForCPB = 1 => on or ForCPB = 0 => off VERTICAL DIFFUSION HEATING  
/
```

Experimentos **ForCLW=0** Exclui o impacto da radiação de onda longa

```
&ForcingControl  
  ForCLW = 0 , !ForCLW = 1 => on or ForCLW = 0 => off LONG WAVE RADIATIVE HEATING  
  ForCSW = 1 , !ForCSW = 1 => on or ForCSW = 0 => off SHORT WAVE RADIATIVE HEATING  
  ForCDC = 1 , !ForCDC = 1 => on or ForCDC = 0 => off CONVECTIVE LATENT HEATING  
  ForCLG = 1 , !ForCLG = 1 => on or ForCLG = 0 => off LARGE SCALE LATENT HEATING  
  ForCSC = 1 , !ForCSC = 1 => on or ForCSC = 0 => off SHALLOW CONVECTIVE HEATING  
  ForCPB = 1 , !ForCPB = 1 => on or ForCPB = 0 => off VERTICAL DIFFUSION HEATING  
/
```



Use o namelist e veja a contribuição de cada parametrização modificando as forçantes (ForCLW, ForcSW, ForcDC, ForcPB)

OBS: Lembre-se que o namelist “**namelist.input**” deve estar no diretório que você irá executar o modelo

Experimentos **ForcSW=0** **Exclui o impacto da radiação de onda curta**

```
&ForcingControl  
  ForCLW = 1 , !ForCLW = 1 => on or ForCLW = 0 => off LONG WAVE RADIATIVE HEATING  
  ForcSW = 0 , !ForcSW = 1 => on or ForcSW = 0 => off SHORT WAVE RADIATIVE HEATING  
  ForcDC = 1 , !ForcDC = 1 => on or ForcDC = 0 => off CONVECTIVE LATENT HEATING  
  ForCLG = 1 , !ForCLG = 1 => on or ForCLG = 0 => off LARGE SCALE LATENT HEATING  
  ForcSC = 1 , !ForcSC = 1 => on or ForcSC = 0 => off SHALLOW CONVECTIVE HEATING  
  ForcPB = 1 , !ForcPB = 1 => on or ForcPB = 0 => off VERTICAL DIFFUSION HEATING  
/  

```



Use o namelist e veja a contribuição de cada parametrização modificando as forçantes (ForCLW, ForcSW, ForcDC, ForcPB)

OBS: Lembre-se que o namelist “**namelist.input**” deve estar no diretório que você irá executar o modelo

Experimentos **ForcSW=0** **Exclui o impacto da convecção**

```
&ForcingControl
```

```
ForCLW = 1 , !ForCLW = 1 => on or ForCLW = 0 => off LONG WAVE RADIATIVE HEATING  
ForcSW = 1 , !ForcSW = 1 => on or ForcSW = 0 => off SHORT WAVE RADIATIVE HEATING  
ForcDC = 0 , !ForcDC = 1 => on or ForcDC = 0 => off CONVECTIVE LATENT HEATING  
ForCLG = 1 , !ForCLG = 1 => on or ForCLG = 0 => off LARGE SCALE LATENT HEATING  
ForcSC = 1 , !ForcSC = 1 => on or ForcSC = 0 => off SHALLOW CONVECTIVE HEATING  
ForcPB = 1 , !ForcPB = 1 => on or ForcPB = 0 => off VERTICAL DIFFUSION HEATING
```

```
/
```



Use o namelist e veja a contribuição de cada parametrização modificando as forçantes (ForCLW, ForcSW, ForcDC, ForcPB)

OBS: Lembre-se que o namelist “**namelist.input**” deve estar no diretório que você irá executar o modelo

Experimentos **ForcSW=0** **Exclui o impacto da convecção**

```
&ForcingControl
```

```
ForCLW = 1 , !ForCLW = 1 => on or ForCLW = 0 => off LONG WAVE RADIATIVE HEATING  
ForcSW = 1 , !ForcSW = 1 => on or ForcSW = 0 => off SHORT WAVE RADIATIVE HEATING  
ForcDC = 1 , !ForcDC = 1 => on or ForcDC = 0 => off CONVECTIVE LATENT HEATING  
ForCLG = 1 , !ForCLG = 1 => on or ForCLG = 0 => off LARGE SCALE LATENT HEATING  
ForcSC = 1 , !ForcSC = 1 => on or ForcSC = 0 => off SHALLOW CONVECTIVE HEATING  
ForcPB = 0 , !ForcPB = 1 => on or ForcPB = 0 => off VERTICAL DIFFUSION HEATING
```

```
/
```



Use o namelist e veja a contribuição de cada parametrização modificando as forçantes (ForCLW, ForcSW, ForcDC, ForcPB)

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Experimentos **ForcSW=0** **Exclui o impacto da convecção**

```
&ForcingControl  
  ForCLW = 1 , !ForCLW = 1 => on or ForCLW = 0 => off LONG WAVE RADIATIVE HEATING  
  ForcSW = 1 , !ForcSW = 1 => on or ForcSW = 0 => off SHORT WAVE RADIATIVE HEATING  
  ForcDC = 1 , !ForcDC = 1 => on or ForcDC = 0 => off CONVECTIVE LATENT HEATING  
  ForCLG = 1 , !ForCLG = 1 => on or ForCLG = 0 => off LARGE SCALE LATENT HEATING  
  ForcSC = 1 , !ForcSC = 1 => on or ForcSC = 0 => off SHALLOW CONVECTIVE HEATING  
  ForcPB = 0 , !ForcPB = 1 => on or ForcPB = 0 => off VERTICAL DIFFUSION HEATING  
/  

```



Objetivo final

Discuta o que acontece com as variáveis prognósticas U, V, T e Q em relação ao experimento com todas as forçantes ligadas